

SHADOWFOX

DATA ANALYST INTERNSHIP

Retail Sales Analysis Dashboard

Project Documentation Report — Beginner Level

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Project Title	Retail Sales Analysis Dashboard
Internship Level	Beginner
Tool Used	Microsoft Excel
Dataset	Madhav E-Commerce Sales Dataset
Program	ShadowFox Data Analyst Internship

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1. Project Overview

This project was completed as part of the ShadowFox Data Analyst Internship, Beginner Level track. It focuses on analyzing retail/e-commerce sales data using Microsoft Excel, in line with the internship's requirement to convert raw structured business data into useful spreadsheet-based findings and a simple dashboard.

The objective was to take raw order and transaction details and turn them into a clean, analysis-ready dataset, and then use standard spreadsheet analysis techniques in Excel to understand:

- Overall sales performance
- Profitability
- Monthly sales movement
- Category performance
- Sub-category/product performance
- Regional sales performance
- Payment-mode behavior

The final output of the project is a simple Excel dashboard that brings together key performance indicators (KPIs), charts, business insights, and recommendations in one place.

2. Dataset Used

The project uses the Madhav E-Commerce Sales Dataset, which is provided as two related files — an Orders file and a Details file. Each file captures a different part of the transaction: the Orders file holds customer and location information, while the Details file holds the product and payment information for each order.

Orders File

Contains 500 order records with the following fields:

- Order ID
- Order Date
- Customer Name
- State
- City

Details File

Contains 1,500 detail records with the following fields:

- Order ID
- Amount
- Profit
- Quantity

- Category
- Sub-Category
- Payment Mode

The two files are connected to each other using the common field Order ID.

3. Data Integration

Since the Orders and Details files each held only part of the picture, the first practical step was to bring them together into one working table using Order ID as the common key. The resulting combined table was named Sales_Data.

The merged Sales_Data table contains the following fields:

- Order ID
- Order Date
- Customer Name
- State
- City
- Amount
- Profit
- Quantity
- Category
- Sub-Category
- Payment Mode

The final merged dataset contains 1,500 rows, matching the number of detail records, since each detail record represents one product line within an order.

An important point to note here is that one order can contain more than one product or detail line. Because of this, the same Order ID can appear more than once in the merged dataset. These repeated Order IDs are not duplicate errors — they are valid, expected entries at the transaction (detail) level, and they were treated as such throughout the analysis.

4. Data Cleaning and Preparation

Before any analysis could be trusted, the merged data needed to be checked and standardized. The following steps were carried out:

- Data types were reviewed and corrected across all fields.
- Order ID was kept as Text, since it is an identifier and not a value to be calculated on.
- Order Date was formatted as Date, to allow correct chronological sorting and month-wise grouping.
- Customer Name, State, City, Category, Sub-Category, and Payment Mode were treated as Text fields.

- Amount and Profit were treated as numeric/decimal values, since they are used in calculations.
- Quantity was treated as a whole number.
- The dataset was checked for missing values.
- Actual duplicate detail rows (identical records) were checked and treated separately from repeated Order IDs, which, as explained above, are a normal part of the transaction-level structure.
- Column names were cleaned for readability — for example, CustomerName was renamed to Customer Name, and PaymentMode was renamed to Payment Mode.

The cleaned and combined analysis table produced from this process was named SalesData, forming the base for all further analysis in the workbook.

5. Excel Workbook Structure

The workbook was organized into separate sheets so that raw data, working data, and final output are kept clearly apart. This also makes the workbook easier to review and follow.

Sheet	Purpose
Orders	Original order-level information as received.
Details	Original transaction/detail-level information as received.
Sales_Data	Merged, analysis-ready dataset combining Orders and Details.
Analysis	Summary metrics, monthly analysis, category analysis, sub-category analysis, state analysis, payment-mode analysis, and business insights.
Dashboard	Final presentation layer containing KPIs, charts, insights, and recommendations.

6. Key Summary Metrics

A set of core summary metrics was calculated on the Analysis sheet to give a quick, high-level view of overall business performance:

- Total Sales
- Total Profit
- Total Quantity Sold
- Total Orders
- Average Order Value

Total Orders was calculated based on the count of unique Order IDs, rather than simply counting the number of rows in Sales_Data. This was necessary because, as noted earlier, an Order ID can repeat across multiple detail rows, so a simple row count would overstate the actual number of orders.

Average Order Value was calculated as:

$$\text{Average Order Value} = \text{Total Sales} \div \text{Total Orders}$$

These metrics form the KPI cards later shown on the Dashboard sheet, giving stakeholders an immediate summary before they look into the detailed charts.

7. Monthly Sales Analysis

To understand how sales moved over time, Order Date was used to group transactions by Year-Month. Sales, Profit, and Quantity were each summarized at this monthly level, and a monthly trend chart was created to visualize strong and weak periods across the year.

Month	Observation	Approx. Sales
January 2018	Highest sales month	₹61,632
July 2018	Lowest sales month	₹12,966

Business meaning: Sales clearly varied across months rather than staying flat. This kind of monthly pattern is useful for a retail business because it can support decisions around inventory planning, promotional timing, and overall sales planning for similar periods in the future.

8. Category Analysis

Transactions were grouped by Category and analyzed using Sales, Profit, and Quantity, to see which product categories were driving the business.

Category	Metric	Approx. Value
Electronics	Highest Sales	₹166,267
Clothing	Highest Profit	₹13,325

Sales and profit were deliberately looked at separately, because a category with high revenue is not necessarily the category with the highest profitability. In this dataset, Electronics led on sales, while Clothing led on profit — a distinction that matters when deciding where to focus business attention.

9. Sub-Category Analysis

The category-level view was expanded further to the Sub-Category level, again analyzing Sales, Profit, and Quantity, to get a more specific picture of which products were performing best.

Printers emerged as the highest-selling sub-category, with approximately ₹59,252 in sales. This points to a strong, well-performing product area that could be considered for focused inventory planning and possible cross-selling with related products.

10. State-Level Analysis

Using the location fields (State and City) from the Orders data, sales and profit were analyzed at the state level to identify which regions were contributing the most to overall business performance.

Maharashtra recorded the highest state-level sales, at approximately ₹102,498. This kind of regional breakdown helps identify strong markets that may deserve continued or increased business focus.

11. Payment Mode Analysis

Payment Mode was analyzed using Sales and Quantity to understand customer payment preferences across the dataset.

Cash on Delivery (COD) was the most-used payment mode based on quantity, accounting for approximately 2,456 units. Understanding payment behavior in this way can support planning around customer service and payment-related operations.

12. Dashboard Design

A dedicated Dashboard sheet was built as the final, presentation-ready layer of the workbook. It was designed to be simple and easy to read at a glance, in keeping with the Beginner-level scope of this project.

KPI Cards

- Total Sales
- Total Profit
- Total Orders
- Average Order Value

Charts

- Monthly Sales Trend
- Sales by Category
- Sales by Payment Mode
- Sales by Sub-Category

Business Insights Section

A concise summary of the strongest findings from the analysis, written for quick reading.

Recommended Actions Section

Short, practical business recommendations drawn directly from the findings.

The dashboard was intentionally kept simple, uncluttered, and easy to navigate, since the goal of a Beginner-level project is to demonstrate clear, correct spreadsheet analysis rather than complex, advanced-analytics-style reporting.



Figure 1. Final Retail Sales Analysis Dashboard developed in Microsoft Excel

13. Business Insights

1. Sales Peak

January 2018 recorded the highest monthly sales at approximately ₹61,632, while July 2018 recorded the lowest at approximately ₹12,966.

2. Category Leader

Electronics generated the highest sales at approximately ₹166,267.

3. Product Hotspot

Printers were the strongest sub-category, with approximately ₹59,252 in sales.

4. Market & Payment

Maharashtra led state-level sales, while COD was the most-used payment mode based on quantity.

14. Business Recommendations

1. Plan promotions and inventory around stronger sales periods, based on the monthly trend observed.
2. Maintain sufficient stock and promotional attention for high-performing categories such as Electronics.
3. Review strong sub-category demand, especially Printers, for possible cross-selling and product-bundle opportunities.
4. Consider regional and payment preferences when planning sales initiatives and customer-facing strategies.

15. Business Value

Overall, this project helps a business to:

- Understand overall sales performance
- Identify strong and weak periods
- Compare categories and sub-categories
- Identify strong regional markets
- Understand payment behavior
- Support basic inventory, promotional, and sales-planning decisions

16. Testing and Validation

Before finalizing the workbook, the following checks were carried out:

- Correct data types across all fields
- Correct row structure in Sales_Data
- Correct handling of repeated Order IDs
- Accurate KPI calculations
- Correct chronological monthly ordering
- Correct category/sub-category grouping
- Correct chart source ranges
- Correct currency formatting
- Correct ₹ symbol display
- Clear dashboard layout
- Readable chart titles and labels

17. Limitations

- The analysis is descriptive and based on the available dataset only.
- It identifies patterns and performance differences but does not establish causation.
- Historical data beyond the available period was not available for longer-term trend analysis.
- The project focuses on summary-level business analysis rather than predictive modeling.

18. Future Enhancements

- Add more historical periods for stronger trend analysis
- Add customer-level segmentation

- Add region-wise drill-down
- Add interactive Excel slicers if appropriate
- Add product-level profitability analysis
- Explore forecasting in a future, more advanced project

19. Conclusion

This project took raw e-commerce order and transaction data and, through a structured process of integration, cleaning, and spreadsheet analysis in Excel, turned it into a clean analysis table and a simple, readable dashboard. The resulting KPIs, monthly and category-level trends, and regional and payment-mode findings provide a practical, business-oriented view of sales performance. As a Beginner-level project, it demonstrates the core skills of preparing real-world business data and interpreting it into meaningful, actionable insights.

20. Final Project Workflow

